

SOFIA ODEH

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Spanish (EU) citizen | Eligible to work in Switzerland and the EU without sponsorship | Available January 2027

EDUCATION

Northeastern University – College of Engineering

Boston, MA

Bachelor of Science in Mechanical Engineering, Minor in Astrophysics | GPA: 3.8

Dec 2026

Coursework: System Analysis and Control, Fluid Mechanics, Thermal Systems, Materials Selection (Grad), Modern Physics
Dean's List | PEAK Award | Joseph M. Tucci Scholarship | SHPE | Student Alumni Ambassador

Teaching Assistant: Measurements and Analysis (ME 4505/4506) – grading and office hours, 10 hrs/week

Fall 2026

RELEVANT EXPERIENCE

LineSpect – Imaging & Optics Startup

San Francisco, CA

Mechanical and Optical Engineer

Jan – Apr 2025

- Designed and prototyped precision optical housings and mounts using Autodesk Inventor and SLA/FDM 3D printing
- Supported optical system optimization, gaining exposure to Zemax lens design and light-sheet imaging workflows
- Characterized optical systems with ImageJ, analyzing field of view, depth of field, and sheet thickness
- Repaired electronics and soldered SMT components while maintaining and calibrating 3D printers

Aramaki Lab – Northeastern University

Boston, MA

Research Collaborator – Gamma-Ray and Antimatter Survey (GRAMS)

Sep 2024 – Present

- Analyzed 1M+ cosmic muon events using Python, achieving Landau distribution fitting with MPV at 7.1 mV
- Constructed 4x4 scintillator arrays with Teflon wrapping and SiPM coupling for enhanced particle detection
- Engineered mechanical fixtures for TPC positioning in liquid argon tanks with precise alignment constraints
- Operated CAEN digitizer systems to characterize timing coincidences and identify cosmic muon trajectories

Lutron Electronics – Lighting & Control Systems

Boston, MA

Mechanical Engineering Intern

Jan – Jun 2024

- Designed and qualified tooling with detailed drawing packets, focusing on manufacturability
- Performed rigorous tolerance analysis to ensure precise fit between PCBs and enclosures
- Designed extensively in Creo Parametric, coupled with SLA and FDM 3D printing for rapid prototyping
- Led cross-functional design reviews, aligning tool quality with production and operational requirements

Silicon Synapse Lab – Northeastern University

Boston, MA

Research Assistant

May – Aug 2023

- Collaborated on COBRA, a bio-inspired snake-like exploration robot for crater observation
- Integrated Raspberry Pi sensors via ROS for improved obstacle detection and autonomous navigation

SEDS – Students for the Exploration and Development of Space

Boston, MA

Mechanical Lead – NEST (Networked Exploration Swarmed Toolkit)

Apr 2024 – Dec 2025

- Led a team developing networked mini-COBRA modules for coordinated lunar exploration
- Spearheaded CAD redesign and prototyping for untethered battery integration and autonomous operation
- Executed 3D printing, assembly, and testing using SLA/FDM techniques to improve module scalability

LEADERSHIP & CREATIVE PROJECTS

Art of Inquiry Podcast

Boston, MA

Founder & Host

Sep 2025 – Present

- Created and host a podcast exploring where creativity and science intersect, interviewing leading researchers
- Full production pipeline: recording, editing (Audacity/Premiere Pro), brand identity design, Spotify distribution

Boston Clothing Outreach

Boston, MA

Founder

Apr 2024 – Present

- Founded initiative combining clothing design with social impact; donated ~1,800 lbs of clothing to local centers
- Designed and produced original garments from donated materials, launching a small clothing collection

NU SCI Magazine – Newsletter Founder

Sep 2023 – 2025

- Pioneered a weekly email newsletter presenting science in a non-traditional, curiosity-driven format

SKILLS

Languages: English (Native), Spanish (Native), Italian (Intermediate)

Engineering: SolidWorks, Autodesk Inventor, Creo Parametric, AutoCAD, MATLAB, Python, Exposure to Zemax, Machining, SLA/FDM 3D Printing, Laser Cutting, Soldering/SMT, ROS, FEA (in progress, Fall 2026)

Analysis: ImageJ, Data Analysis (Python), Tolerance Analysis

Certifications: DELE C1 (Diplomas of Spanish as a Foreign Language), Excel Essentials

Interests: Rock & Blues, Design, Swing Dance, Astronomy, Poetry & Writing